CS & IT ENGINEERING

Digital Logic

Combinational Circuits

Lecture No. 02

By- Aditya sir

Recap of Previous Lecture

Topic

K-maps

Duality

Intro to Combinational ckts.

Topics to be Covered

Topic

Combinational ckts

→ Misc Concepts + Ques

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram

@ADITYASIR_PW

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

* Combinational Circuit :

→ 1) 1 bit magnitude Comparator :

5 Step framework for designing Combinational ckt :-

Step 1 :- Find the no. of i/ps and o/ps.

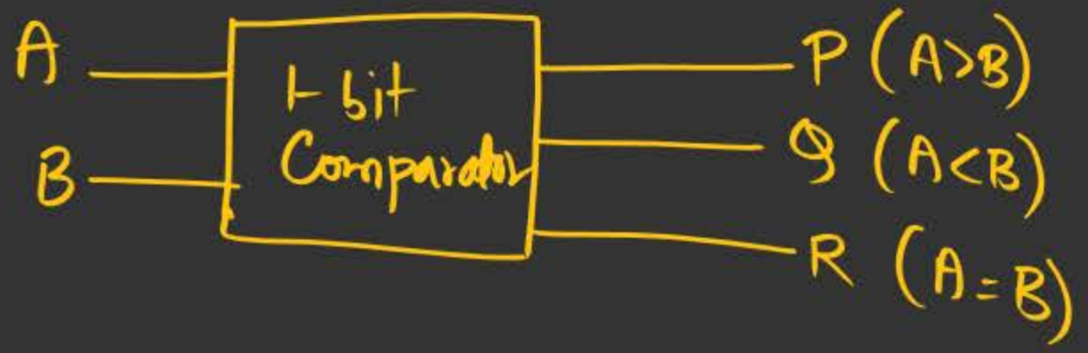
Step 2 :- Create the truth table.

Step 3 :- Write down the logical expression.

Step 4 :- Minimize the logical expression.

Step 5 :- Design the circuit / Hardware impletn.

Summary of 1-bit Comparator:-



$$P \ (A>B) = A\cdot\overline{B}$$

$$Q \ (A<B) = \overline{A}\cdot B$$

$$R \ (A=B) = \overline{A}\cdot\overline{B} + A\cdot B$$

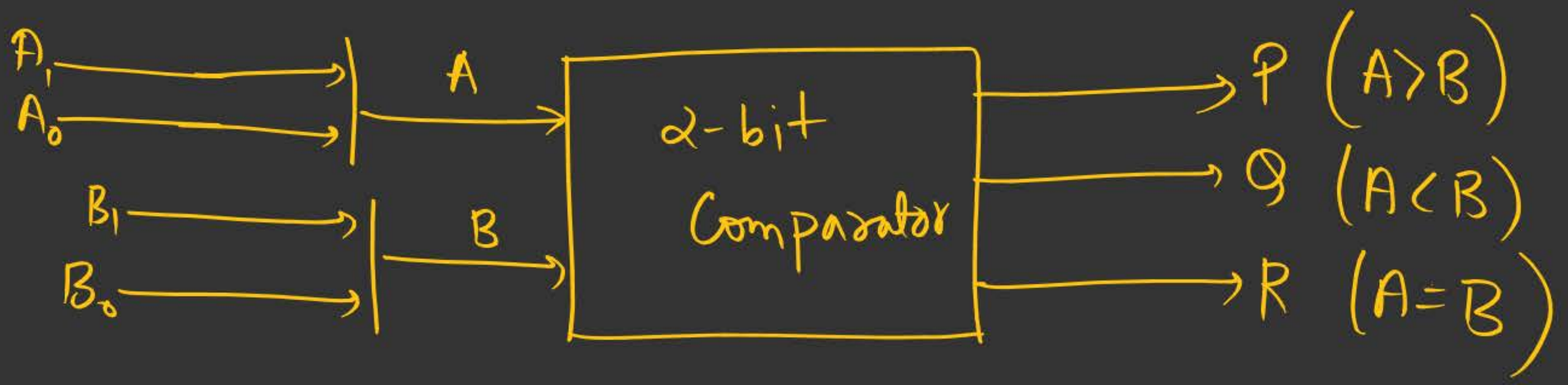
$$= A \odot B$$

2) 2-bit Magnitude Comparator :-

i/p : A & B

$$\underbrace{A_1 A_0} \quad \underbrace{B_1 B_0}$$

Step1 : I/p & O/p : ✓



MSB

MSB

A_1	A_0	B_1	B_0
1	_	0	_

eg:- 79 > 42

79 75

eg:-

A: A_1	A: A_0	B: B_1	B: B_0	
0	0	0	1	→ $A < B$
1	0	0	1	→ $A > B$
1	1	1	1	→ $A = B$

Step 2

True - 1
False - 0

Decimal	A: A_1	A: A_0	B: B_1	B: B_0	(A>B) P	(A<B) Q	(A=B) R
0	0	0	0	0	0	0	1
1	0	0	0	1	0	1	0
2	0	0	1	0	0	1	0
3	0	0	1	1	0	1	0
4	0	1	0	0	1	0	0
5	0	1	0	1	0	0	1
6	0	1	1	0	0	1	0
7	0	1	1	1	0	1	0
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	0	1
11	1	0	1	1	0	1	0
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	0	1

Step 3 :- Logical Expression :-

1) $P(A>B) : \sum m(4,8,9,12,13,14)$

2) $Q(A<B) : \sum m(1,2,3,6,7,11)$

3) $R(A=B) : \sum m(0,5,10,15)$

Step 4:- Minimization

For P $\Rightarrow \sum m(4, 8, 9, 12, 13, 14)$

$A > B$

$A_1A_0 \backslash B_1B_0$	$\bar{B}_1\bar{B}_0$	$\bar{B}_1B_0$	B_1B_0	$B_1\bar{B}_0$
$\bar{A}_1\bar{A}_0$	0	1	3	2
$\bar{A}_1A_0$	1 (4)	5	7	6
A_1A_0	1 (12)	1 (13)	15	1 (14)
$A_1\bar{A}_0$	1 (8)	1 (9)	11	10

$A_0\bar{B}_1\bar{B}_0$

$A_1A_0\bar{B}_0$

$A_1\bar{B}_1$

Minimized SOP:

$$f = A_0\bar{B}_1\bar{B}_0 + A_1A_0\bar{B}_0 + A_1\bar{B}_1$$

$$= A_0\bar{B}_0(\bar{B}_1 + A_1) + A_1\bar{B}_1$$

Step 4:- Minimization

For P $\Rightarrow \sum m(4, 8, 9, 12, 13, 14)$

$A > B$

$A_1A_0 \backslash B_1B_0$	$\bar{B}_1\bar{B}_0$	$\bar{B}_1 B_0$	$B_1 B_0$	$B_1\bar{B}_0$
$\bar{A}_1\bar{A}_0$	0	1	3	2
$\bar{A}_1 A_0$	1 (4)	5	7	6
$A_1 A_0$	1 (12)	1 (13)	15	1 (14)
$A_1\bar{A}_0$	1 (8)	1 (9)	11	10

$\bar{A}_1 A_0 \bar{B}_1 \bar{B}_0$

$A_1 A_0 B_1 \bar{B}_0$

$A_1\bar{B}_1$

★ Semi-minimized

$$f = A_1\bar{B}_1 + \bar{A}_1 A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 B_1 \bar{B}_0$$

$$= A_1\bar{B}_1 + (\bar{A}_1\bar{B}_1 + A_1 B_1) \cdot A_0\bar{B}_0$$

$$= A_1\bar{B}_1 + (A_1 \odot B_1) \cdot A_0\bar{B}_0$$

B > A

for Q :- $\sum m(1,2,3,6,7,11)$

$A_1A_0 \backslash B_1B_0$	$\bar B_1 \bar B_0$	$\bar B_1 B_0$	$B_1 B_0$	$B_1 \bar B_0$
$\bar A_1 \bar A_0$	0	1 1	1 3	1 2
$\bar A_1 A_0$	4	5	1 7	1 6
$A_1 A_0$	12	13	15	14
$A_1 \bar A_0$	8	9	1 11	10

$\bar A_1 \bar A_0 B_0$

$\bar A_1 B_1$

$\bar A_0 B_1 B_0$

minimized SOP

$$f = \bar A_1 B_1 + \bar A_1 \bar A_0 B_0 + \bar A_0 B_1 B_0$$

$$= \bar A_1 B_1 + (\bar A_1 + B_1) \cdot \bar A_0 B_0$$

for G :- $\sum m(1,2,3,6,7,11)$

$A_1A_0 \backslash B_1B_0$	$\bar{B}_1\bar{B}_0$	$\bar{B}_1B_0$	B_1B_0	$B_1\bar{B}_0$
$\bar{A}_1\bar{A}_0$	0	1 (1)	1 (3)	1 (2)
$\bar{A}_1A_0$	4	5	1 (7)	1 (6)
A_1A_0	12	13	15	14
$A_1\bar{A}_0$	8	9	1 (11)	10

$\bar{A}_1\bar{A}_0\bar{B}_1B_0$

$\bar{A}_1B_1$

$A_1\bar{A}_0B_1B_0$

B > A

Semi-minimized SOP

$$F = \bar{A}_1B_1 + \bar{A}_1\bar{A}_0\bar{B}_1B_0 + A_1\bar{A}_0B_1B_0$$

$$= \bar{A}_1B_1 + (\bar{A}_1\bar{B}_1 + A_1B_1)\cdot\bar{A}_0B_0$$

$$= \bar{A}_1B_1 + (A_1 \odot B_1)\cdot\bar{A}_0B_0$$

for R: $\Sigma m(0,5,10,15)$

$A_1A_0 \backslash B_1B_0$	$\bar B_1\bar B_0$	$\bar B_1 B_0$	B_1B_0	$B_1\bar B_0$
$\bar A_1\bar A_0$	1 (0)	(1)	(3)	(2)
$\bar A_1 A_0$	(4)	1 (5)	(7)	(6)
A_1A_0	(12)	(13)	1 (15)	(14)
$A_1\bar A_0$	(8)	(9)	(11)	1 (10)

$\bar A_1\bar A_0\bar B_1\bar B_0$

$\bar A_1 A_0\bar B_1 B_0$

$A_1A_0B_1B_0$

$A_1\bar A_0 B_1\bar B_0$

$$f = \bar A_1\bar A_0\bar B_1\bar B_0 + \bar A_1 A_0\bar B_1 B_0 + A_1A_0B_1B_0 + A_1\bar A_0B_1\bar B_0$$

$$= \bar A_1\bar B_1(\bar A_0\bar B_0 + A_0B_0) + A_1B_1(A_0B_0 + \bar A_0\bar B_0)$$

$$= (A_0B_0 + \bar A_0\bar B_0)\cdot(A_1B_1 + \bar A_1\bar B_1)$$

$$= (A_0 \odot B_0)\cdot(A_1 \odot B_1)$$

$$= (A_1 \odot B_1)(A_0 \odot B_0)$$

1-bit

$(A \odot B)$

52 52

A → A_1A_0

B → B_1B_0

Summary:- Semi-Minimised SOP (Imp Shortcut)

1) $P(A>B) \Rightarrow A_1\overline{B_1} + (A_1 \odot B_1)\cdot A_0\overline{B_0}$

$\downarrow$

2 bit

2) $P(A>B) \Longrightarrow A\cdot\overline{B}$

for 1-bit

$A \rightarrow A_1 A_0$ $\quad B \rightarrow B_1 B_0$

7 2 $\quad$ 4 9

7 5 $\quad$ 7 1

2) Q (B>A) $\Rightarrow \overline{A_1} B_1 + (A_1 \odot B_1) \cdot \overline{A_0} \cdot B_0$

2-bit Comp:-

2) Q (B>A)

1-bit $\Rightarrow \overline{A} \cdot B$

$A \rightarrow A_1 A_0$ $B \rightarrow B_1 B_0$

A	B
45	63
45	49

eg:- Shortcut to directly write <u>3-bit Comparator</u>

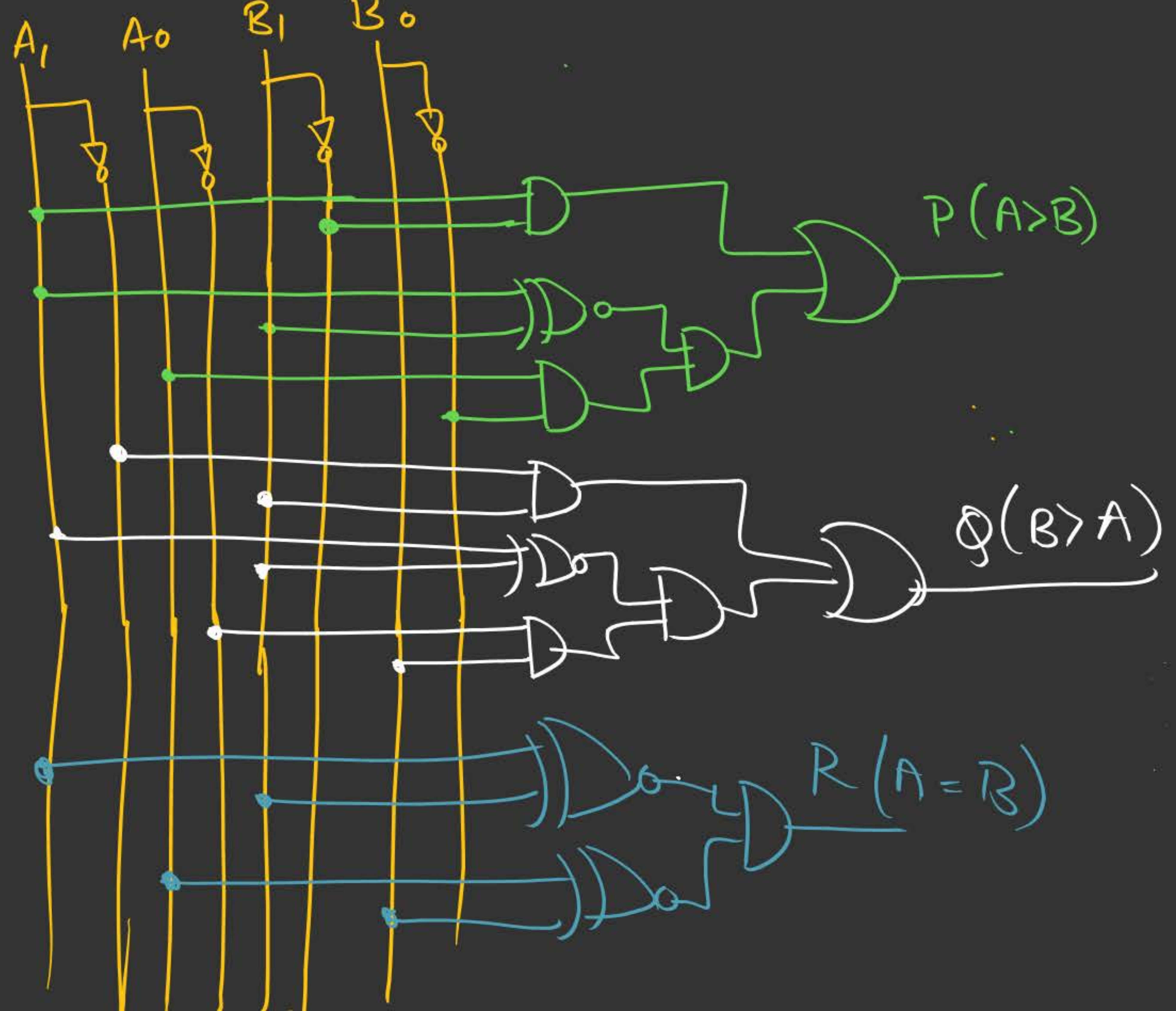
$$\begin{array}{cc} A & B \\ \downarrow & \downarrow \\ A_2 A_1 A_0 & B_2 B_1 B_0 \end{array}$$

1) $P\,(A>B) \Rightarrow A_2 \cdot \overline{B}_2 + (A_2 \odot B_2) \cdot A_1 \overline{B}_1 + (A_2 \odot B_2) \cdot (A_1 \odot B_1) \cdot A_0 \overline{B}_0$.

2) $Q\,(B>A) \Rightarrow \overline{A}_2 B_2 + (A_2 \odot B_2) \cdot \overline{A}_1 B_1 + (A_2 \odot B_2)(A_1 \odot B_1) \cdot \overline{A}_0 B_0$

3) $R\,(A=B) \Rightarrow (A_2 \odot B_2)(A_1 \odot B_1) \cdot (A_0 \odot B_0)$

Step 5 :- Hardware design :



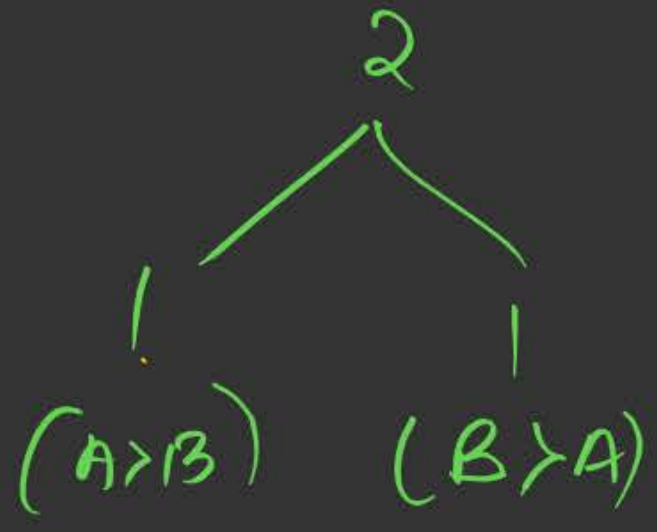
Properties / Observation (Comparators)

1) **1-bit Comparator:**

→ Total Combinations/Condition = 4

→ Equal Combinations (R = 1) = 2

→ Unequal Combinations (R = 0) = (4 − 2) = 2



→ #Greater Comb^n = #lesser Comb^n = 1

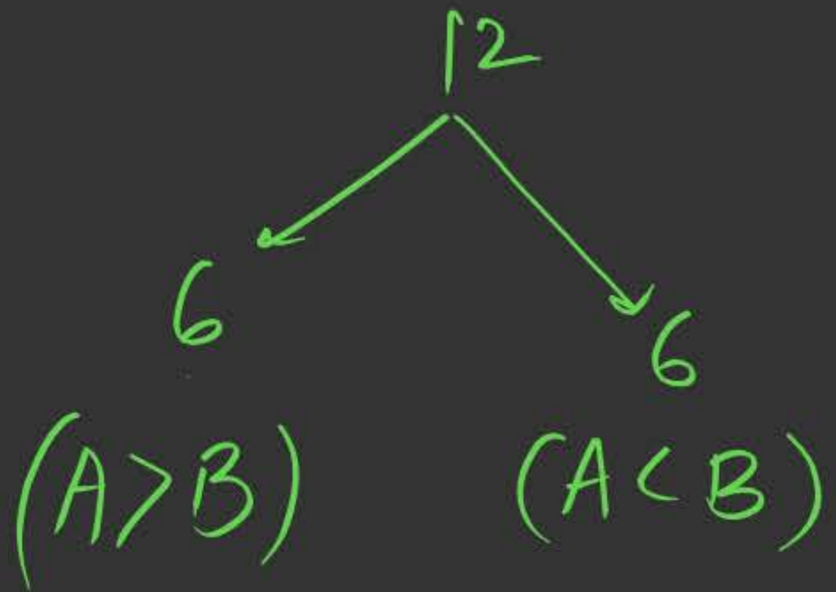
2) **2-bit Comparator :-**

→ Total Combi = 16

→ Equal Combi = 4

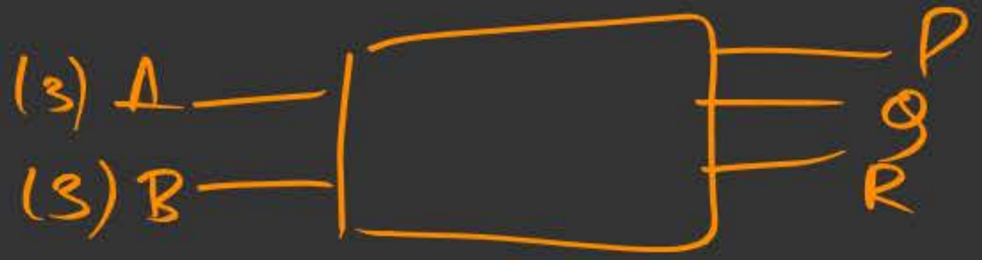
→ Unequal " = (16-4) = 12

→ # Greater = # lesser = 6



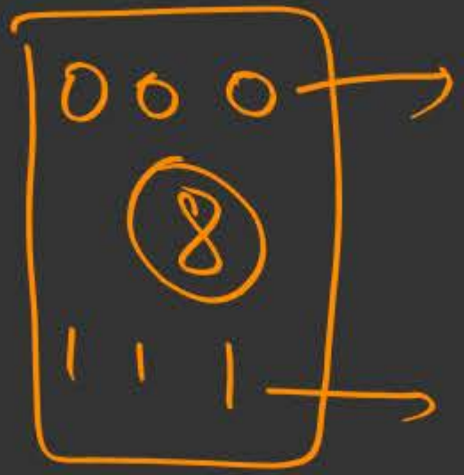
$2^6 = 64$

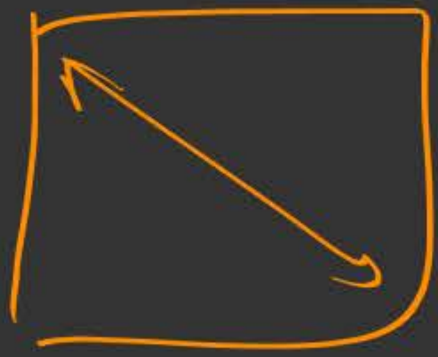
3) 3-bit Comparator :



→ Total Combi = 64

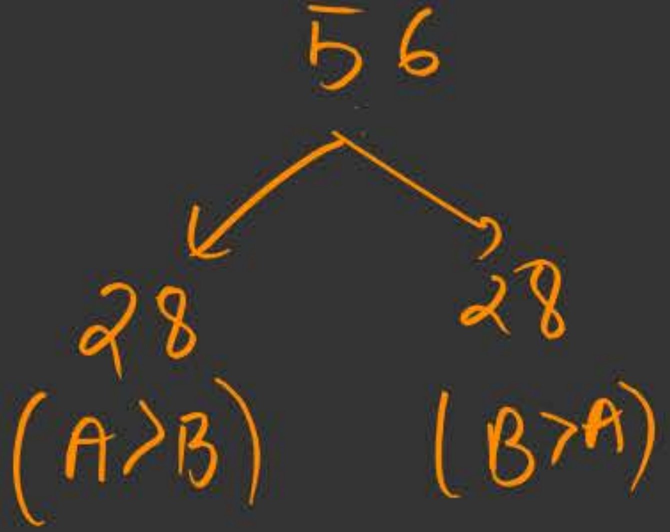
→ Equal Combi = 8





→ Unequal Combi = (64 − 8) = 56

→ # greater = # lesser = $\frac{56}{2} = 28$



Generalised (V.V. Imp Shortcut)

n-bit Comparator

4 16 64

1) Total conditions/Combinations = $2^{(2n)}$

2) Equal Combi = 2^n

3) Unequal Combi = $\left(2^{(2n)} - 2^n\right)$

4) # Greater Combi = # lesser Combi = $\dfrac{\left(2^{2n} - 2^n\right)}{2}$

12, 13 → No digital logic class

→ DPP - 2 → Solve

→ Practice $\boxed{\text{entire K-map}}$ (2 times)

→ Solve PYQs on Ch-1 & Ch-2 content.

2 mins Summary

→ Comparator

(1-bit
2-bit
3-bit)

obs

→ Expression logic
(Shortcut)

THANK - YOU